

Question 32

Bias in AI refers to a systematic error in an AI system that produces unfairly skewed or discriminatory outcomes for example favouring certain age, race, gender etc.

When an AI model is biased, it means that its decisions are influenced by errors in the training data or in the design of the algorithm

An example is when a hiring AI tool favours males because it was trained on data that contained more males. The consequences of this flaw is that of economic disadvantage and unfairness and potential legal liabilities

Question 33

The differences between Data Analyst, Data Scientist, Data Engineer & AI Engineer

Role	Core focus	Skills	Tools
Data Analyst	- Interprets historical & present data to find trends, answer business questions & create visual reports for stakeholders	- SQL, Statistical analysis, problem-solving, data visualization, business acumen	- Tableau - Power BI - Excel, SQL - Google sheets
Data Scientist	- Uses advanced maths, statistics and machine learning to predict future actions and solve problems	- Machine learning - Data modelling - Python, R, Statistics - Hypothesis testing	Python, R, SQL Jupyter notebook, Tensorflow Scikit-learn, Databricks
Data Engineer	- Builds & maintains data infrastructure, storage & pipelines to ensure data is clean, accessible & smooth flow	- Software engineering - ETL, Database architecture - Distributed systems	Apache spark SQL, Hadoop Google Bigquery
AI Engineer	- Builds scalable AI applications that learn and improve over time for e.g chatbots	- Software engineering - APIs, Prompt engineering - LLM orchestration, MLOps	Python, OpenAI API LangChain, Docker Pytorch, Kubernetes